

A Proven and Tested Clustering Technique in Modeling Energy Systems

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Abstract

We present a revolutionary binary clustering technique that promises to redefine the field of binary data analysis. Applied successfully on a renewable energy system of biomass gasification in fluidized bed reactors, this novel approach offers a paradigm shift in the way binary data is clustered and interpreted. Traditional binary clustering methods often struggle to capture the unique characteristics and patterns inherent in data, limiting their effectiveness in real-world applications. Our proposed technique leverages the eigenvalues and eigenvectors of a characteristic matrix to address the challenges specific to pattern discovery. More than 200 data sets of biomass gasification in fluidized bed were collected featuring a wide range of operating condition and fuel types. An axiom-based reasoning was used to develop a multiphase statistical pathway needed as a precondition to effectively quantify the entanglements of different important factors in the process. Specifically, by creating an interconnected chain of analysis based on trigonometric functions, geometric projections, and design of a statistical inference tool utilizing neural network units, multiple partial measures of associations between biomass constituents, and operating condition were effectively consolidated and embedded in a single characteristic matrix that consequently led to detection of monotonic relationships for prediction of carbon conversion efficiency and product gas yield over the largest applicable range of all the influential parameters of the process, namely, temperature, air equivalent ratio, steam to biomass ratio, and type of fuel. In part of our methodology, we introduce a novel technique for obtaining a dynamical property value for stationary objects, based on a “specific computational time” of an “abstract mechanical operation on characteristics matrices”. The specific computational time (sct) showed excellent capability in capturing the non-equilibrium factor of the process which itself was function of several interrelated variables. By integrating state-of-the-art algorithms with robust mathematical foundations, we have developed a method that improves upon existing approaches in terms of accuracy, ease of implementation, and speed of convergence. Key to our technique is allowing the inclusion of prior knowledge or expert guidance to guide the clustering process. This not only enhances the precision of the clustering results but also facilitates the extraction of meaningful insights from data. Furthermore, our approach harnesses the power of advanced neural networks, enabling the automatic discovery of complex patterns and hierarchical structures within data. This technique holds great promise to accurately identify clusters, reveal hidden relationships, and provide valuable insights for decision-making across various domains.

Keywords: Renewable Energy Systems, Multivariate Compositional Analysis, Pattern Discovery, Biomass Gasification, Fluidized Bed

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1 Introduction

2 Literature Review

3 General Methodology

3.1 Input Consolidation: Construction of Representative Group

The goal is to moderate the extreme numerical fluctuations in parameters that describe events, while still preserving the distinct identity of each event. It's crucial to strike the right balance between overall homogeneity and the independence of individual cases. To satisfy the homogeneity requirement, it's essential to start from a common baseline. Thus, we start the transformation with a circle having an area of one, akin to a standard probability density function, which is non-negative throughout and integrates to one over its entire domain.

In the last step of "representative" preparation we aim to recuperate its overall "uniformity" which may have been weakened due to multiplexity of steps taken after the equity circle. For analysis of inherently noisy data, diverse, and subject to various source of variations, a preprocessing technique as part of feature engineering is to construct a set of congruent inputs to neural network and achieve ultra fast convergence to an optimal solution during training.

3.2 Multifaceted Sensitivity Analysis: Gaining Holistic Insights

In practical implementations, the slope parameter allows for a neuron to be more sensitive (steep slope) or less sensitive (shallow slope) to changes in its input. Adjusting this slope can be useful in certain training scenarios or when trying to achieve specific neuron behaviors. (*multilateralpath*)

Here, since we are dealing with a convoluted object, it does not seem economical to perform rigorous structural analysis of each case, therefore, we base our judgement on the "general principle" of equilibrium for stationary objects. To be stationary all of the forces on the object must be in equilibrium, the resultant force and moment vectors must both be zero. From structural point of view, the number of equilibrium conditions required to solve different force systems can be summarized as following:

4 Application

4.1 Research Objective

The overall objective of this study is to develop a general model for biomass gasification in fluidized bed with aim of predicting the gasifier output (i.e. H₂, CO, CO₂, CH₄, Carbon Conversion Efficiency and Product Gas Yield) by adopting effective strategies to cope with complex dynamics of the process, limitation of experimental data, and lack of proper characterization of solid fuel. We start with applying a sequence of machine learning predictive data analytics to extract features and patterns related to many properties of biomass that in previous models that have not been taken into account for, namely the structural component of biomass (cellulose, hemicellulose, lignin), the proximate analysis of biomass fuel (volatile matter, fixed carbon, ash content), This study aims to extend the generality of the model through a systematic quantification of collective impact of different critical factors such as temperature (T), equivalent air ratio (ER), steam to biomass ratio (SBR) in play to make it applicable to a wide range of biomass classes. From practical point of view, it is desirable to maintain the computational time of model reasonably short (i.e. close to the black box models). Such a model will allow one to optimize the blend of biomass fuel and operating condition to meet desirable specifications for various applications of interest including power, hydrogen, methanol and Fischer-Tropsch hydrocarbons.

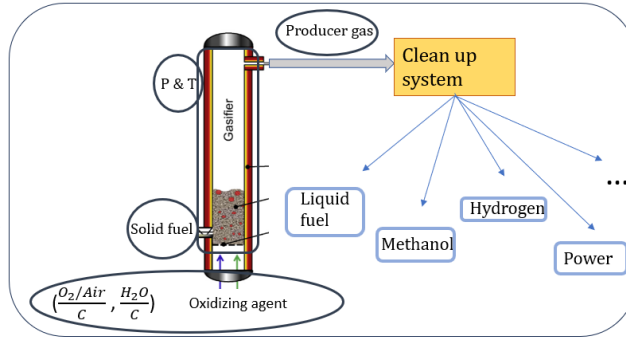


Figure 1: process overview

4.2 Biomass Characterization

The starting point of analysis is the biomass. In summary, the characterization of biomass with respect to its application in gasification process can be divided into three categories:

1. Types of analyses that are well known, usually investigated in experiments, and are directly used as inputs of majority of models (i.e. elemental composition).
2. Types of analyses that their impacts are recognized but not investigated in experiments as extensively as type 1, and therefore their consideration in models are limited (i.e. proximate analysis and structural components).
3. The unknown characteristics (or extremely variable features) of biomass fuel that results in presence of an irreducible amount of uncertainty about the feedstock composition and structure (i.e. biomass ash).

Biomass can be classified in several groups such as woody, herbaceous, aquatic, animal waste, municipal solid waste, and etc. These groups themselves can be classified to other sub groups based on different criteria such as origin, organic component, minerals and mode of element occurrence

4.2.1 Canonical Correlation Analysis

We've devised a method that employs CCA analysis to seamlessly integrate various biomass properties. This procedure has multiple stages, beginning with a preconditioning step. In this initial stage, we establish an intrinsic connection within each property group. To do this, we select the variable with the smallest variation coefficient in each group to serve as the standard for normalization. Specifically, we use Carbon for the elemental group, Fixed Carbon for the proximate group, and Hemicellulose for the structural group. Subsequently, we apply a logarithmic transformation to these ratios, ensuring a direct internal correspondence between the elements of each group.

Finally, to interconnect the elemental, proximate, and structural groups, we undertake two distinct CCA analyses. This number – two – represents the fewest connections needed to ensure all three groups are interlinked.

$$\begin{aligned} \bullet E_1 &= Ln(\frac{C^{1.45}}{O^{0.17}N^{0.63}S^{0.65}}), P_1 = Ln(\frac{FC^{0.52}VM^{0.38}}{A^{0.9}}), \\ \bullet E_2 &= Ln(\frac{H^{0.88}}{C^{0.31}N^{0.23}S^{0.34}}), S_2 = Ln(\frac{Hem^{0.17}Cell^{0.48}}{Lig^{0.65}}), \end{aligned}$$

Figure 2: compact set

4.2.2 Geometric Transformation

Construction of geometric objects to reduce the vector of $[E1, E2, P1, S2]$ to $[d, K]$ which are basically a distance and an area and calculates as follows:

The value of the $EPS = [E1, E2, S1, P1]$ is calculated for all the datasets. Two triangles, $\Delta E1E2S1$ with centroid G_s and $\Delta E1E2P1$ with centroid G_p , are constructed.

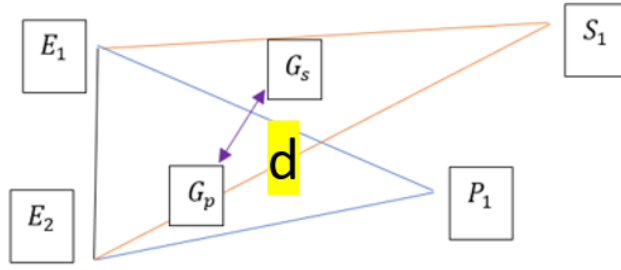
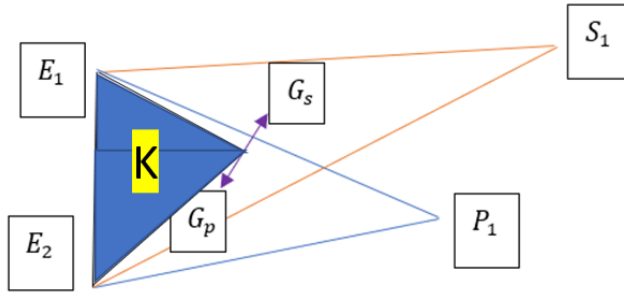


Figure 3: triangle

We calculate the distance d between two centroids and use it as a relative measure to assess how the structural component of biomass and its proximate analysis are positioned relative to its elemental composition. Therefore, G_{ps} is our initial critical parameter in this context.

The second parameter chosen is the area of the triangle formed by the base $E1E2$ and the midpoint of the $G_p - G_s$ line.



Let K be the area of the triangle with vertices given by (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) :

$$K = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \quad (1)$$

4.2.3 Construction of Representative Group

The total area is partitioned into four sections, reflecting the four sources that influence the gasifier’s output: the overall biomass and three distinct operating parameters—temperature, equivalent air ratio, and steam to biomass ratio. To account for potential combined effects, we’ve intentionally defined each section with overlaps. To achieve optimal results, it’s crucial to iteratively refine the distributions, ensuring that deep domain knowledge about the specific processes is integrated. Below are the guiding principles used in this context:

- (a) **Significance of Parameters:** The prominence given to a particular parameter in shaping an area correlates with its importance, as drawn from domain knowledge. For instance, while temperature has its distinct section, it also serves a supplementary function in other sections.
- (b) **Attribute of Randomness:** The design ensures a degree of randomness in the distribution. By introducing randomness at an early stage, the model is fortified against high uncertainty.
- (c) **Bounds on Section Shares:** To prevent any section from becoming overly dominant, we’ve defined upper and lower limits on each section’s share. For example, we avoided omitting the SBR section to ensure consistent segmentation, avoiding fluctuation between three and four sections.
- (d) **Inclusiveness Hierarchy:** We’ve structured an internal hierarchy among the sections based on the scope of their inclusiveness.

Given the scale of our input set $[d, k, T, ER, SBR]$, we have adopted the subsequent formulation and configuration to satisfy our criteria. The coefficients were iteratively adjusted to better align with the typical composition of syngas, our target for prediction.

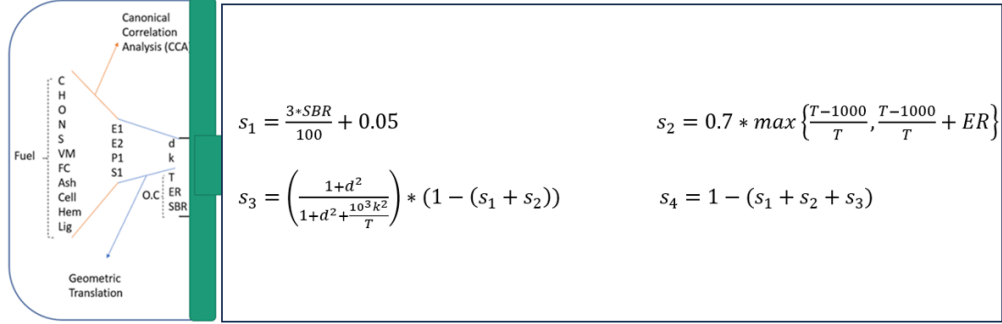


Figure 5: feature engineering

We have adopted the well-established two-tiered representation system. In this structure, each section is represented both by its "population or size" and independently of its size. Specifically, we:

- Preserve the size property, denoted by the area of the circular sector.
- Allocate "exactly one full quadrant" to every section, irrespective of its size.

From a geometric perspective, uniformity among partitions and the perpendicular lines dividing the four regions are essential features. The product of the ratios of altitudes in right triangles exhibited a strong 72 percent correlation with the same variable derived from the output set. This robust correlation underscores the effectiveness of the measures implemented thus far in constructing profoundly informative attributes. This criterion has remained central in the pursuit of exceptionally influential predictors. The output set has been calculated in similar fashion, with the product gas composition instead of S1-S4.

- $s_1 = \frac{3 \cdot SBR}{100} + 0.05$
- $s_2 = 0.7 \cdot \max \left\{ \frac{T-1000}{T}, \frac{T-1000}{T} + ER \right\}$
- $s_3 = \frac{(1+d^2)}{(1+d^2+10^3 k^2/T)} \cdot (1 - (s_1 + s_2))$
- $s_4 = 1 - (s_1 + s_2 + s_3)$
- $R_j = \sum_{i=1}^4 r_i$
- $S_j = \frac{l_1 l_2}{2} + \frac{l_2 l_3}{2} + \frac{l_3 l_4}{2} + \frac{l_4 l_1}{2}$
- $V_j = \min \left(273 + \frac{v_1}{16} + \frac{v_2}{28} + \frac{v_3}{2} + \frac{v_4}{44}, 940 \right)$
- $L_j = \frac{l_1}{l_4}$

- $N_j = \frac{1}{2} \left(\frac{L_j \cdot V_j}{T_j \cdot S_j \cdot \ln \ln(R_{max} - R_j)} + 12 \right) \cdot \frac{2.3 \cdot R}{F}$ article amsmath

(a) $f_1 = N_j + 2e^{-N_j}$

(b) $f_2 = e^{-N_j} \cdot f_2$

(c) $f_3 = f_3 + f_2$

(d) $f_4 = f_3 + f_2$

(e) $f_5 = f_4 + f_3$

(f) $f_6 = f_5 + f_4$

(g) $f_7 = f_6 + f_5$

(h) $f_8 = f_7 + f_6$

(i) $f_9 = f_8 + f_7$

(j) $f_{10} = f_9 + f_8$

(k) $f_{11} = f_{10} + f_9$

This process renders two integer sequences for input and output sets. Rationale: Preprocessing data plays a pivotal role in alleviating the computational complexity and enhancing the efficiency of various machine learning and data analysis tasks. By carefully crafting preprocessing pipelines, data can be transformed and refined in ways that make subsequent computations less burdensome and more effective. Data preprocessing involves a series of steps such as normalization, scaling, feature selection, and handling missing values. These steps contribute to reducing the dynamic range of features, bringing them to a common scale, and removing irrelevant or noisy information. It should be noted that carbon conversion efficiency (CCE) and product gas yield (Nm3/kg) are the output variables that should be predicted by the model, and not included as part of the event embedded in the sequences.

article array

16	26	42	62	111
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Table 1: F_A

18	29	47	76	123
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Table 2: F_B

34	55	89	145	234
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Table 3: F_AB

4.3 Design of Neural Network as Computational Labs

The subsequent phase involves crafting a mathematical framework tailored to function as a computational laboratory. This entails deriving inferences about the nature of an inquiry based on its performance along an arbitrary path. The fundamental principles that underlie our statistical methodology encompass the axioms of a multifaceted trajectory, analysis of responses across multiple scales, and the interplay between predictors and predictees.

The form of trajectory and the successive stages at which performance is tracked significantly influence the quality and quantity of the information we gather. In our investigation, we delineate a well-defined, sequences of steps that commences with Fibonacci sequences, advances through a series of neural networks, and operations, and culminates in a characteristic matrix.

Three neural network units, each employing a different transfer function (linear, hyperbolic tangent sigmoid, and Elliot symmetric sigmoid), were trained using sets of Fibonacci sequences. The configurations for these networks are depicted in Figure 49. The ABT model is highlighted as a representative example to elaborate on the neural network details.

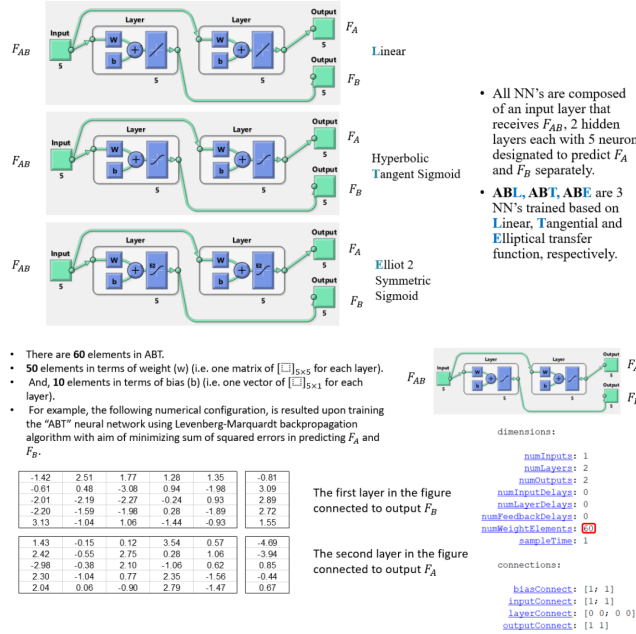


Figure 6: ABT Details

Following the training, the neural network units show properties of a bounded function with respect to the inputs (i.e. F_{AB}), meaning for all the inputs they return vectors of: $F_A = [1, 1, 1, 1, 1]$, $F_B = [1, 1, 1, 1, 1]$. article

This ultra fast converging configuration provides critical information on the holistic state of data distributions. Specifically, the weights and biases calculated upon the training indirectly capture the distribution of input sets with respect to their corresponding output sets.

We will show how the network can form a structure-preserving map of process variables that has a cyclic property across the three different neural network units. This property finds utility in a range of matrix manipulations and calculations, where the reordering of the matrices can lead to simplified expressions, aiding in the analysis of matrix operations, eigenvalues, and determinant evaluations, among other mathematical operations.

Following our strategy to cover broad ranges of possibilities with respect to variation potentials (vertical, horizontal, and cross variations), The scheme is set up as follows: In our study, we conduct a response analysis on the behavior of neural networks (NNs) using three distinct scale factors. The selection of these scales and neural network units (i.e. transfer functions) are dictated by the desired resolution and quantity of information. Depending on specific research needs, there might be occasions that might involve more granular or extensive scales to capture the intricacies of the NN's behavior.

In this study:

- Three different fractions of F_A (i.e., 0.4, 0.7, and 1) are taken into account.
- The input to ABL is in the form of $[0.4 \times F_A, 0.7 \times F_A, F_A]$, while the input to ABT and ABE are in the form of $[\frac{1}{0.4 \times F_A}, \frac{1}{0.7 \times F_A}, \frac{1}{F_A}]$.

To evaluate the behavior of our system, we calculate the relative deviations from two baseline vectors:

$$F_A = [1, 1, 1, 1, 1] \quad \text{and} \quad F_B = [1, 1, 1, 1, 1]$$

For each unique pair generated by the fractions $[0.4, 0.7, 1]$ and the neural network units $[L, T, E]$ —yielding a total of 9 pairs—we perform two distinct sets of calculations as outlined below:

- (a) Changes in F_B with respect to F_A , and F_{AB}
- (b) Changes in F_A with respect to F_B , and F_{AB}

We have established a method for quantification and a scoring system using the following matrix operations. Each matrix is $[5 \times 5]$.

- $[B \times A^T], [dB \times A^T], [dB \times A_{in}^T]$
- $[A \times B^T], [dA \times B^T], [A_{in} \times dA^T]$

In the realm of representation theory, the trace takes on a significant role by serving as the foundation for defining characters within group representations. These characters hold significance across a spectrum of mathematical domains, prominently featuring in the exploration of symmetry groups.

Beyond representation theory, the trace exhibits a profound link to eigenvalues. This property, notable for its mathematical elegance, finds practical relevance in the analysis of diverse systems spanning physics, engineering, and dynamical systems. The trace's ability to encapsulate the collective behavior of eigenvalues renders it an indispensable tool for comprehending the intricate dynamics governing these systems.

Combining the traces of matrices within each group yields a set of scores, as depicted in Figure 8.

		L	T	E	Sum
B to A	0.4	1.05	3.94	3.30	8.29
	0.7	1.67	3.64	3.00	8.31
	1	2.14	3.52	2.90	8.56
A to B	0.4	1.12	-1.28	3.43	3.26
	0.7	1.71	-1.12	2.94	3.54
	1	2.10	-1.04	2.70	3.75

Figure 7: scores

The table summarizes changes across four axioms:

- Path (L, T, E)
- Scale (0.4, 0.7, 1)
- $A \rightarrow B$
- $B \rightarrow A$

Each entry in Table 6 signifies the sum of traces for three matrices within the mentioned groups.

This data originates from the traces of 54 $[5 \times 5]$ matrices, specifically, the outcome of combining 3 scales, 3 paths, and 6 measurements—totaling 54 matrices encompassing 1350 numerical values.

These values likely encapsulate infinitesimal relationships between A and B , all corresponding to a single inquiry.

4.4 Selection of Reference Object

A pivotal consideration inherent to autonomous functional analyses lies in the strategic selection of a reference point from which to infer the potential achievement of quantification of interrelated factors. Equally vital within such analyses is the deliberation on the appropriate "form" and "dimension" (e.g., scalar, vector, involution) that the mathematical construct should ultimately adopt for seamless integration.

Given the intricate nature of the subject under investigation, the feasibility of conducting an exhaustive structural examination for each case becomes impractical. Thus, we rely on a foundational "general principle" derived from the equilibrium of stationary objects to guide our assessments.

The scores in Figure are normalized and an outer product is calculated.

$$- S_{ci} = \frac{1}{2} \times (S_{ABi} \times S_{BAi}^T + S_{BAi} \times S_{ABi}^T)$$

The outer product of two vectors $|S_{AB}\rangle$ and $|S_{BA}\rangle$ results in a characteristic matrix represented by $|S_c\rangle$.

$$\begin{aligned} S_{c1} &= \begin{bmatrix} 1.035 & 1.080 & 1.130 \\ 1.080 & 1.125 & 1.176 \\ 1.130 & 1.176 & 1.229 \end{bmatrix}, & S_{c2} &= \begin{bmatrix} 1.025 & 1.058 & 1.107 \\ 1.058 & 1.089 & 1.139 \\ 1.107 & 1.139 & 1.191 \end{bmatrix} \\ S_{c3} &= \begin{bmatrix} 1.063 & 1.110 & 1.180 \\ 1.110 & 1.156 & 1.226 \\ 1.180 & 1.226 & 1.299 \end{bmatrix}, & S_{c4} &= \begin{bmatrix} 1.086 & 1.143 & 1.222 \\ 1.143 & 1.198 & 1.278 \\ 1.222 & 1.17 & 1.361 \end{bmatrix} \end{aligned}$$

Figure 8: Characteristic Matrix

The characteristic matrix exhibited a common feature among all the 200 cases. The arrows in the following matrix indicate the direction of increase in the numerical values.

$$S_{ci} = \begin{bmatrix} S_{ci}^{11} & S_{ci}^{12} & S_{ci}^{13} \\ S_{ci}^{21} & S_{ci}^{22} & S_{ci}^{23} \\ S_{ci}^{31} & S_{ci}^{32} & S_{ci}^{33} \end{bmatrix}$$

Figure 9: Common Feature

We unveil an innovative approach for extracting dynamic property values from stationary objects. This approach hinges upon a "specific computational time" recorded for an "abstract mechanical operation on characteristic matrices." The specific computational time (sct) showed excellent capability in capturing the non-equilibrium factor of the process which itself was function of several interrelated variables.

The term "abstract mechanical operation on characteristic matrices" refers precisely to the process of transforming the mean product of static vectors (i.e., the characteristic matrix) 3×3 into a vector of 9×1 , and arranging them in ascending order. The interval from instigating this specific procedure to its culmination is meticulously recorded, a span that varies across different instances.

While a consistent environmental backdrop, encompassing factors like device power, prevails in this test, these temporal disparities are postulated to be tethered to the computational rigidity distinctive to each case.

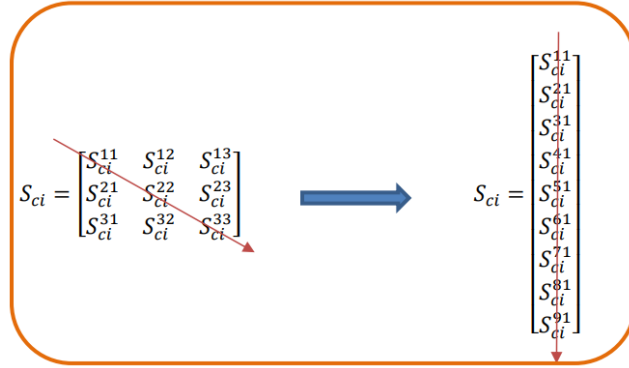


Figure 10: abstract mechanical operation on characteristic matrices

4.5 Complete Elliptic Integral

4.5.1 Choice of Operation

Maintaining a clear focus on the algorithm’s core objective is crucial: the identification of exceptionally influential predictors. One of the viable approaches involves harnessing operations recognized for their capacity to characterize system behavior. As a result, considerations were directed towards filter designs and similar techniques, including various integrals.

A complete “elliptic” integral of the first kind was recognized as the most compatible option. The integration was done through a low-pass filter. Specifically, the Ellipap module in MATLAB was employed. This module returns the zeros, poles, and gain of an order “ n ” elliptic analog low-pass filter prototype, with R_p dB of ripple in the passband, and a stopband R_s dB down from the peak value in the passband (Fig. 38). The zeros and poles are returned in length “ n ” column vectors z and p , and the gain in scalar k . If n is odd, z is of length $n - 1$. The transfer function in factored zero-pole form returns the poles and zeros, and gain of:

$$H(S) = \frac{Z(S)}{P(S)} = k \frac{(s - z_1)(s - z_2) \dots (s - z_N)}{(s - p_1)(s - p_2) \dots (s - p_M)} \quad (70)$$

4.5.2 Arguments of the Integral

Regarding the criteria for feature selection, we investigated potential differentiating factors. Specifically, potential of temporal disparities derived from characteristic matrices to serve as distinguishing features, the Fibonacci sequences where the event was embedded in it, and the absolute temperature which has no substitutes when it comes to modeling energy systems.

- (a) The specific computational time (**sct**)
- (b) The summation of 5 Fibonacci numbers for an inquiry (A) (i.e. the event embedded in the sequences)
- (c) The absolute temperature (T)

This module uses `ellipke` to calculate the “complete elliptic integral” of the first kind and `ellipj` to calculate Jacobi elliptic. It computes the Jacobi elliptic functions using the method of the arithmetic-geometric mean starting with a triplet of numbers.

Initial conditions:

$$a_0 = 1, \quad b_0 = \sqrt{1 - m}, \quad c_0 = \sqrt{m}$$

Recursive equations:

$$a_i = \frac{1}{2} (a_{i-1} + b_{i-1}), b_i = \sqrt{a_{i-1} \times b_{i-1}}, c_i = \frac{1}{2} (a_{i-1} - b_{i-1}).$$

The equation relating $\sin$ and ϕ is:

$$\sin(2\phi_{n-1} - \phi_n) = \frac{c_n}{a_n} \sin(\phi_n)$$

Definitions involving elliptic functions $sn(u)$, $cn(u)$, and $dn(u)$:

$$sn(u) = \sin \phi_0, cn(u) = \cos \phi_0, dn(u) = \sqrt{1 - m \cdot sn(u)^2}.$$

The integral definition for u is:

$$u = \int_0^\phi \frac{d\theta}{\sqrt{1 - m \sin^2 \theta}}$$

Definitions involving elliptic functions $sn(u)$, $cn(u)$, and $dn(u)$:

$$sn(u) = \sin \phi, cn(u) = \cos \phi, dn(u) = \sqrt{1 - m \sin^2 \phi}.$$

Relation between k^2 , m , and $\sin^2 a$:

$$k^2 = m = \sin^2 a$$

In our quest for exceptionally influential predictors, we've explored a viable avenue: leveraging operations known for their capacity to reveal system behavior. This led us to delve into filter designs and similar techniques, including various integrals.

Regarding the criteria for feature selection, we investigated potential differentiating factors. Firstly, we considered the temporal disparities derived from characteristic matrices, aiming to distinguish unique patterns. Secondly, we explored the Fibonacci sequences, which were deeply intertwined with the entire event. Lastly, the absolute temperature stood out due to its unmatched relevance in modeling energy systems. However, the primary rationale behind choosing a complete "elliptic" integral of the first kind stemmed from a significant discovery which will be discussed in the following chapter.

5 Results and Discussion

The zeros and poles are returned in length n column vectors z and p and the gain in scalar k .

	Case 1	Case 2	Case 3	Case 4
z	14×1 complex	16×1 complex	16×1 complex	18×1 complex
p	15×1 complex	16×1 complex	17×1 complex	18×1 complex
k	1.4482×10^{-47}	3.9811×10^{-53}	1.9248×10^{-58}	1.2589×10^{-60}

Output of Integral for Typical Cases

Here it is noteworthy mentioning the integral output lands on solutions with k 's in order of magnitude of the cosmological constants in Einstein's field equations. This essentially serves as a clue:

Since, in general relativity, a fluid solution is an exact solution of the Einstein field equation.

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$$

The value of gains displayed in the third row are denoted with Λ . S is the value computed for a specific case and G is the geometric mean of all cases. $\ln(SG)$ is a dimensionless number that we call the local minima of the algorithm.

The difference between the size of pole and zero vectors is either 1 or 0, which we call the Squarity index. This dimensionless number $\ln(SG)$ provides a quantitative measure correlated with product gas yield. The Squarity index exhibited a monotonic relationship with carbon conversion efficiency. The only distinction lies in the calculation of the Squarity index, which requires determining it for specific bins or intervals since it is a discrete measure.

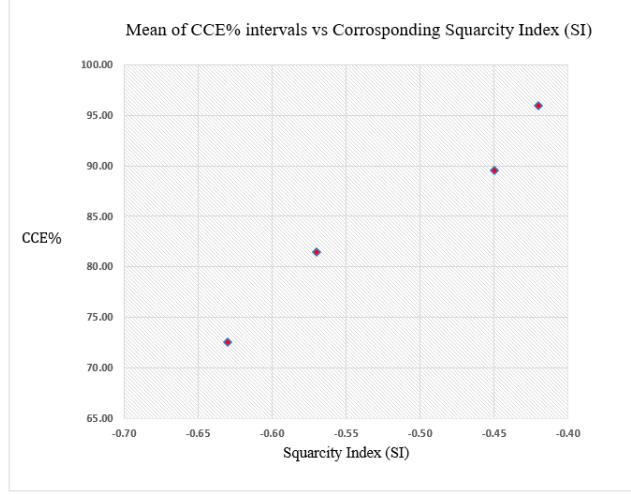


Figure 11: CCE

In numerous exploratory instances, the presence of a strong correlation substantiated the effectiveness of specific feature selection choices. However, at this stage, what truly provided a breakthrough was the exploitation of a particular fact: Interestingly, the output of the integral landed on solutions with gains in the order of magnitude of the cosmological constants in Einstein's field equations. This observation resonated with the insight from general relativity, a fluid solution is an exact solution of the Einstein field equation.

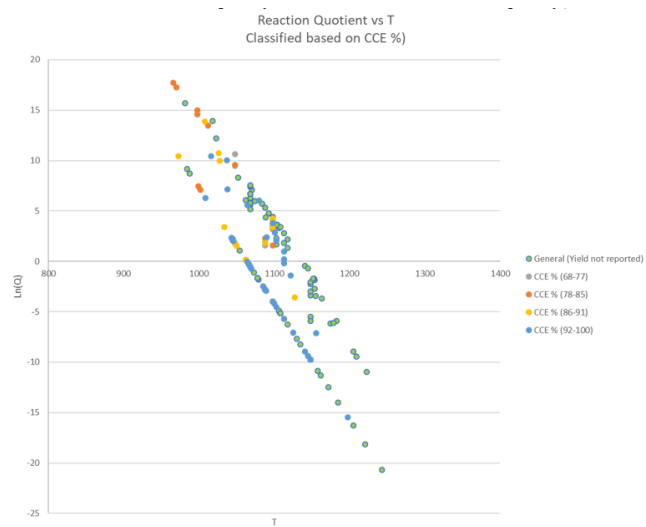


Figure 12: CCE RQ

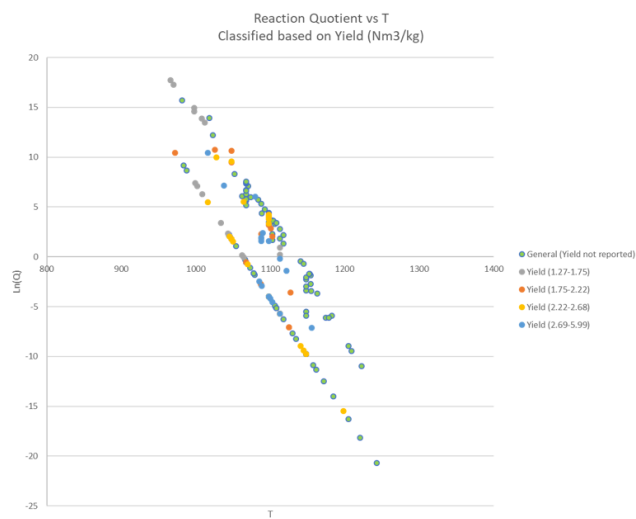


Figure 13: Yield RQ

6 Appendix A - Integral Arguments

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